

Water Treatment Plant Digital Twin

FluxForecast

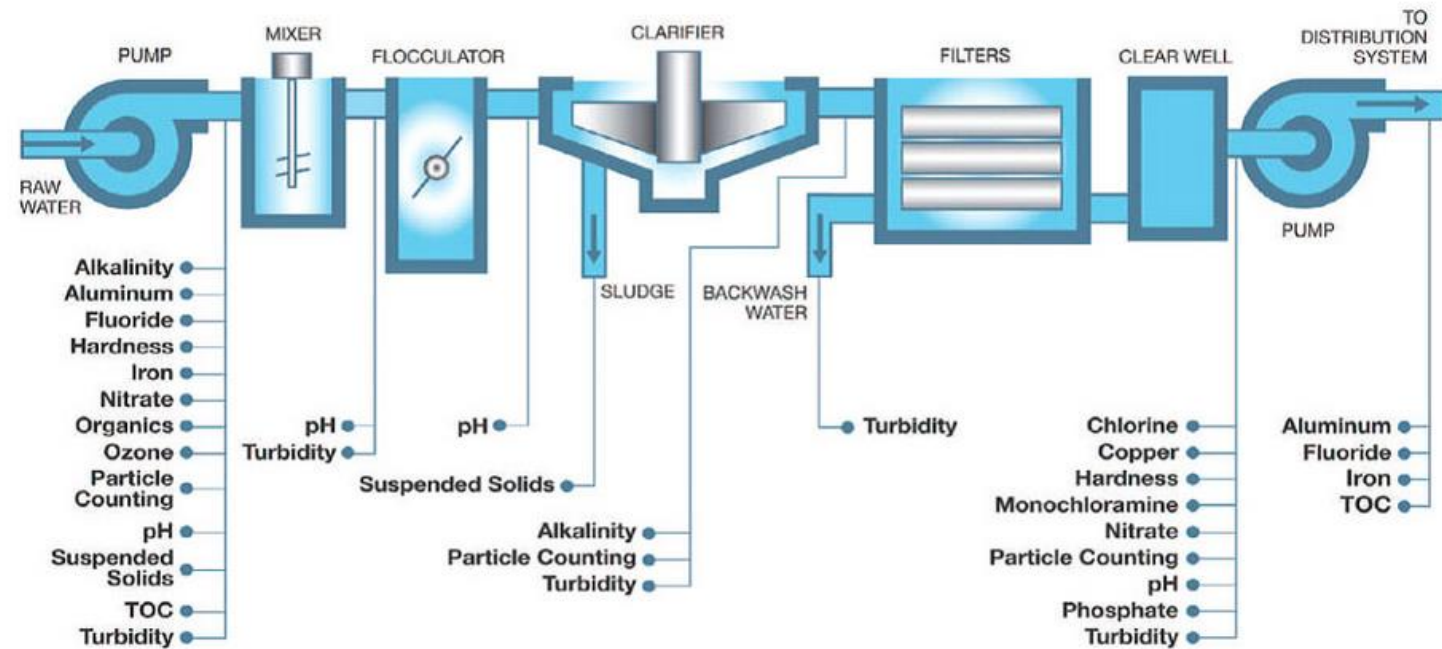
Jason Pereira

Karim Nasser El Harake

Kevin Zhu

Water Treatment Plants (WTPs)

- Conversion of source water to potable water
- **Reactive**, continuous optimization
- Data-rich, but process modelling from first principles is complex.
- AI models can enhance predictive monitoring and process optimization.



Current Opportunities (CA)

- **Few AI Digital Twins**
 - Plant-specific solutions
 - Larger municipalities
- **Similar WTP Prediction Tasks:**
 - CS-BP (Feedforward)¹
 - ARIMA models²
 - Temporal Convolution Network + Long Short Term Memory³
- **Total Operating Cost (2021): \$1.13B CAD**
 - **Opportunity to reduce costs** (i.e. reduce chemical overdosing)



(1) S-H Park and J Koo. *IJCA*. 8(10):135–144, October 2015.

(2) D Shao et al. *Water*. 10(10):1471, October 2018.

(3) Y Xie et al. *Water Research*. 250:121092, February 2024.

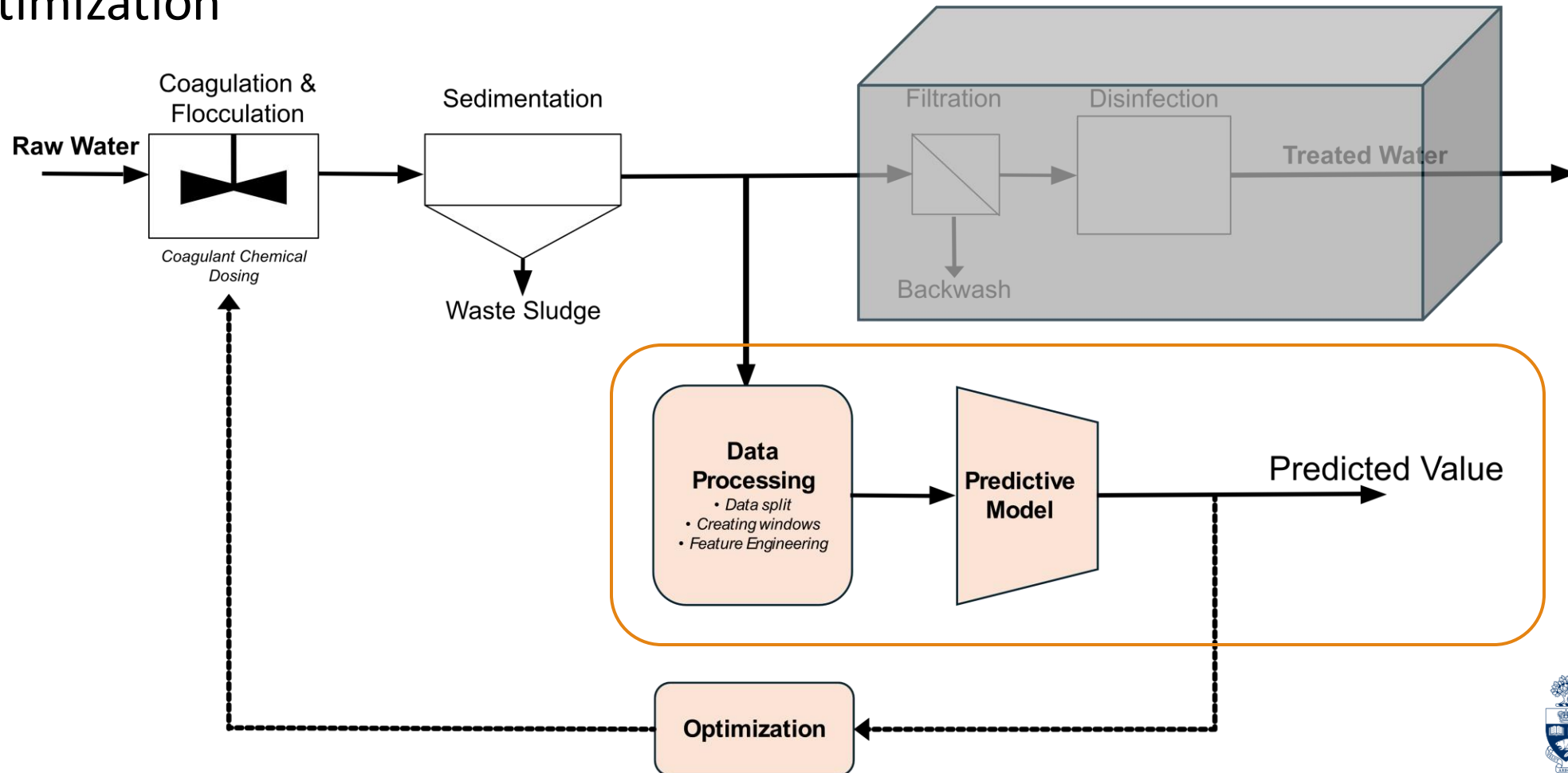
Turbidity

- Measure of **water clarity** due to suspended particles.
- Influenced by pH, dissolved organics, ion concentrations, bacteria, etc.
- Multifaceted **surrogate indicator** of water quality.



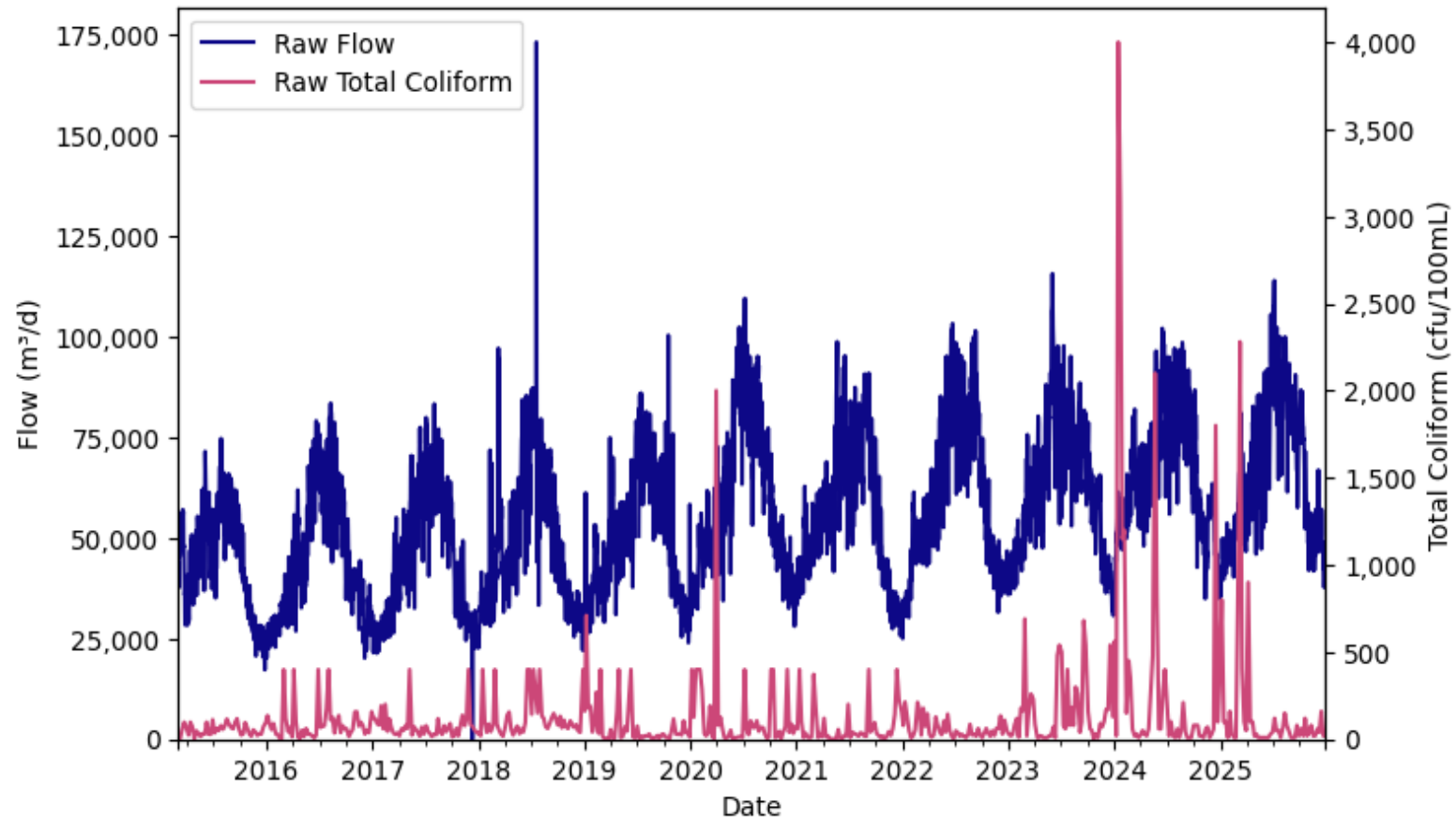
Model Objective

Use raw water and operation data to predict turbidity for process optimization

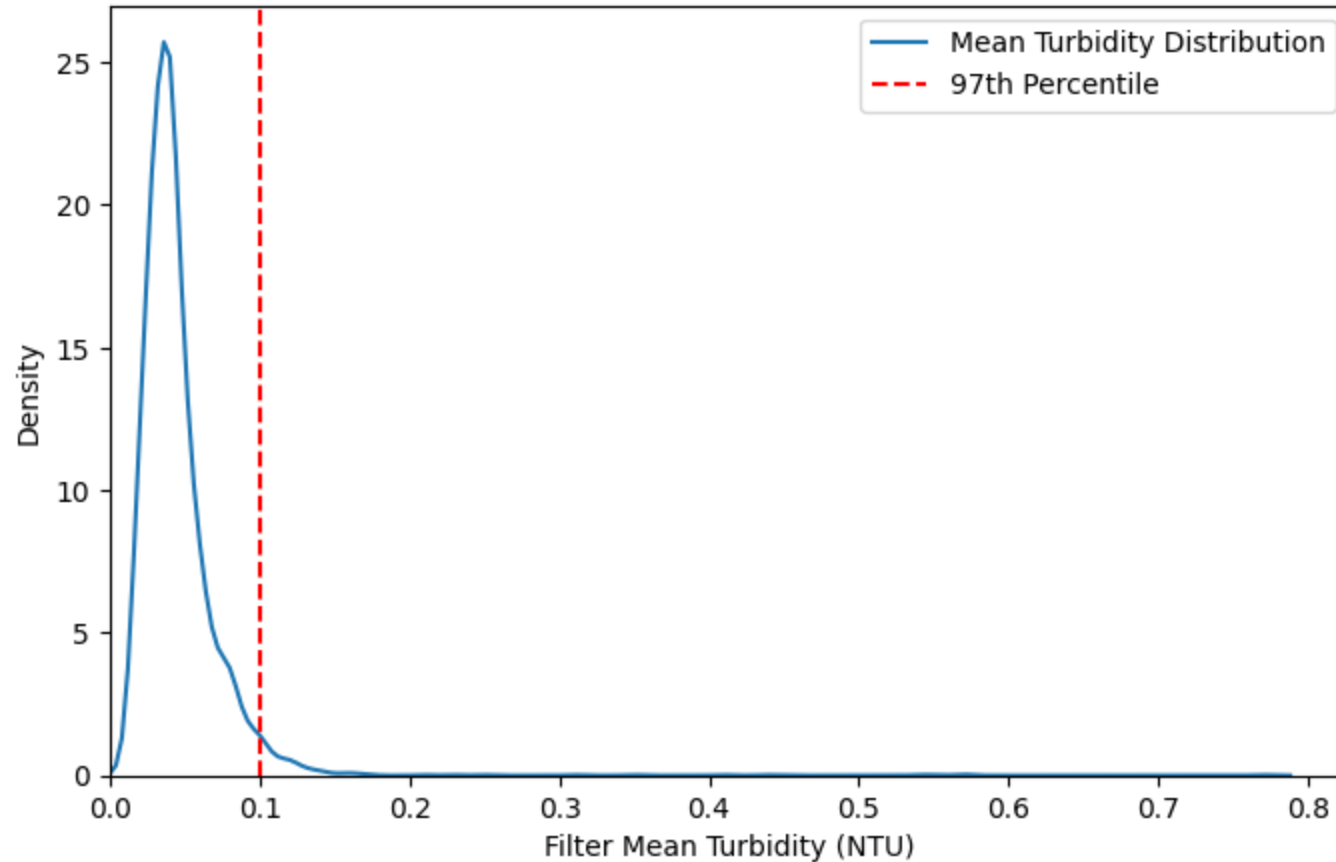


Data Overview

- WTP process data is a unique challenge:
 - Raw water quality affected by **climate** and **weather**.
 - Water demand changes with population growth.
 - Sampling frequency differs for each parameter.

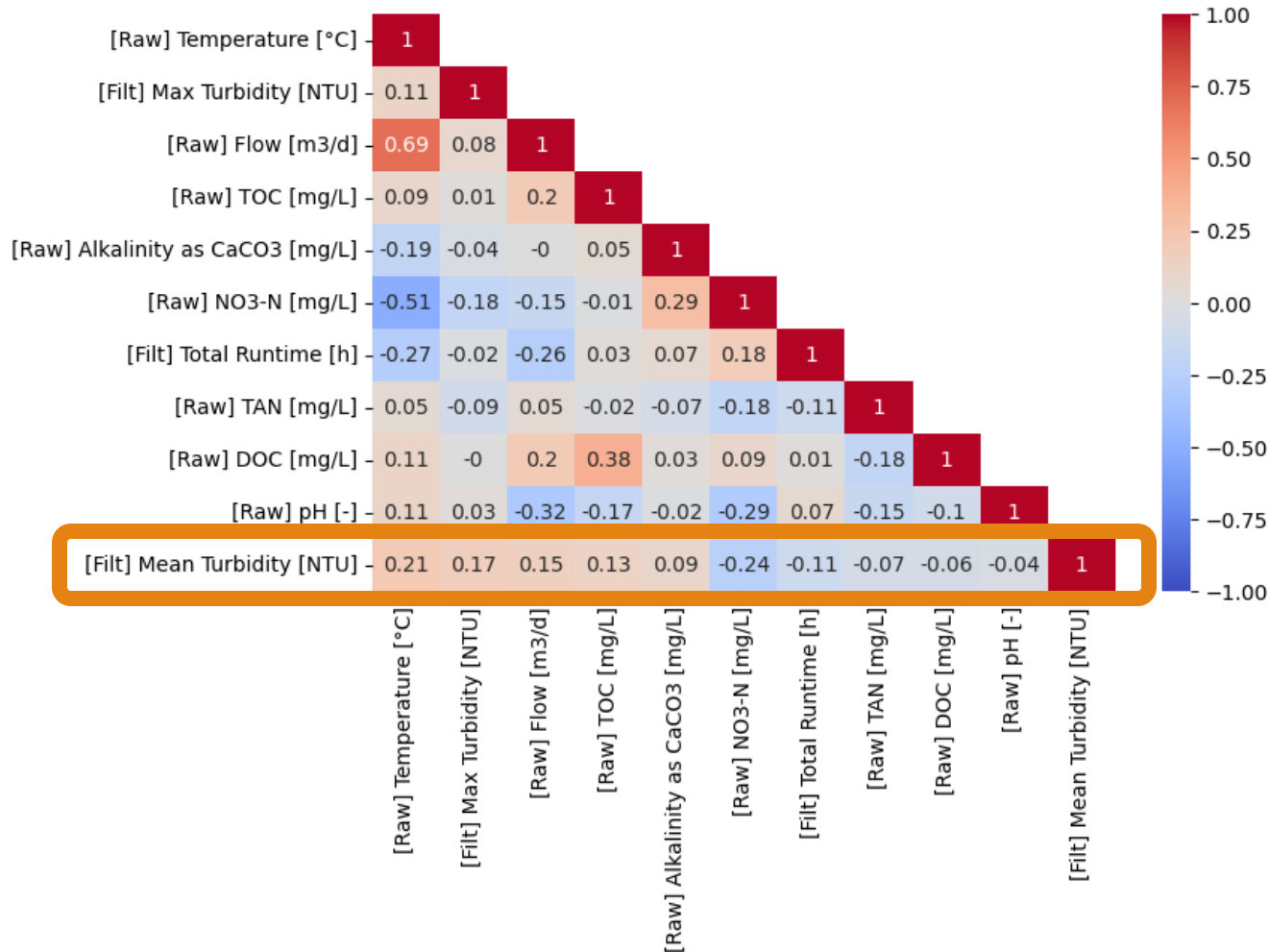


Turbidity Distribution



- **3%** of turbidity data is above 0.1 NTU.
- STD is close to turbidimeter noise (0.031 vs ± 0.02 NTU).
 - Limited discriminatory power and/or high noise.
- **Risk of modelling instability**

Turbidity Correlations



- **Low correlation** with other parameters.
- Some unexpected correlations, which may suggest poor data quality.
 - e.g., Nitrate should trend with turbidity, but correlation is negative!

Data Preprocessing

- WTP data is **irregular and incomplete**, with many variables sampled intermittently (variable frequencies).
- Data exhibits strong **temporal dependence**, requiring time-aware processing/modeling
- A structured preprocessing pipeline was developed to ensure data quality
 - **Window Segmentation** – creating sliding windows across the dataset
 - **Feature Engineering** – creating features, and handling missing values
 - **Model-specific Pre-processing** – e.g., normalization, flattening, etc.

Window Segmentation

- Fixed-length sliding windows are used to capture **temporal dependencies** in WTP data.

Date	[Raw] Flow [m3/d]	[Raw] Alkalinity as CaCO3 [mg/L]	[Raw] Max Turbidity [NTU]	[Raw] Temperature [°C]	[Raw] pH [-]	[Raw] E.Coli [cfu/100mL]	[Chem] Powdered Activated Carbon Dose [mg/L]	[Chem] Alum Dose [mg/L]	[Chem] Polyaluminum Chloride Dose [mg/L]	[Filt] Mean Turbidity [NTU]
2024-12-27	41714		11.7	2.4	7.26		2		27.91	0.05
2024-12-28	41592		14.4	2.9	7.33	40	2.11		30.48	0.03
2024-12-29	35079		6.9	3.1	7.32		2		31.09	0.03
2024-12-30	42547		46.7	4.35	7.29		2.13		30.83	0.04
2024-12-31	40922			3.4	7.26		2.12		30.61	0.04
2025-01-01	35384		35	3.25	7.28		2		29.99	0.05
2025-01-02	39468		81	3	7.37		2		32.49	0.04
2025-01-03	39568		45.4	2.75	7.33		2		30.92	0.04
2025-01-04	42970		52.6	2.05	7.38	50	2		30.87	0.07
2025-01-05	42962		45.9	1.45	7.39				29.84	0.05
2025-01-06	42361	87	29	1.35	7.33		2.07		29.92	0.07
2025-01-07	45533		18.4	1.1	7.35		2.06		29.78	0.1
2025-01-08	49890		16.4	1.1	7.37		2		23.79	0.09

**Not actual data, just for visualization*

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										...

$$Data_{train/val/test} = [(window_1, y_1), (window_2, y_2) \dots]$$

Feature Engineering

- Creating two additional supporting features :
 - **Mask feature** indicates whether a value is observed (1) or missing (0),
 - **Last observed feature** carries forward the most recent measurement
- Preserving temporal information and captures measurement “staleness”.
- Removing NaNs using forward filling
- Data chronologically split

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2024-12-27	41714		0	6	11.7	2.4	7.26		0	8
2024-12-28	41592		0	7	14.4	2.9	7.33	40	1	0
2024-12-29	35079		0	8	6.9	3.1	7.32		0	1
2024-12-30	42547		0	9	46.7	4.35	7.29		0	2
2024-12-31	40922		0	10		3.4	7.26		0	3
2025-01-01	35384		0	11	35	3.25	7.28		0	4
2025-01-02	39468		0	12	81	3	7.37		0	5
2025-01-03	39568		0	13	45.4	2.75	7.33		0	6
2025-01-04	42970		0	14	52.6	2.05	7.38	50	1	0
2025-01-05	42962		0	15	45.9	1.45	7.39		0	1
2025-01-06	42361	87	1	0	29	1.35	7.33		0	2
2025-01-07	45533		0	1	18.4	1.1	7.35		0	3
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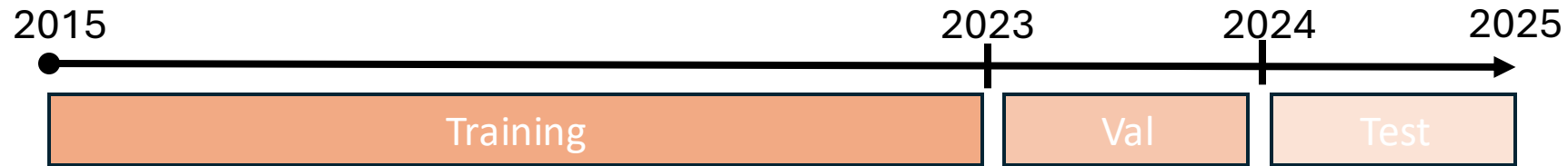
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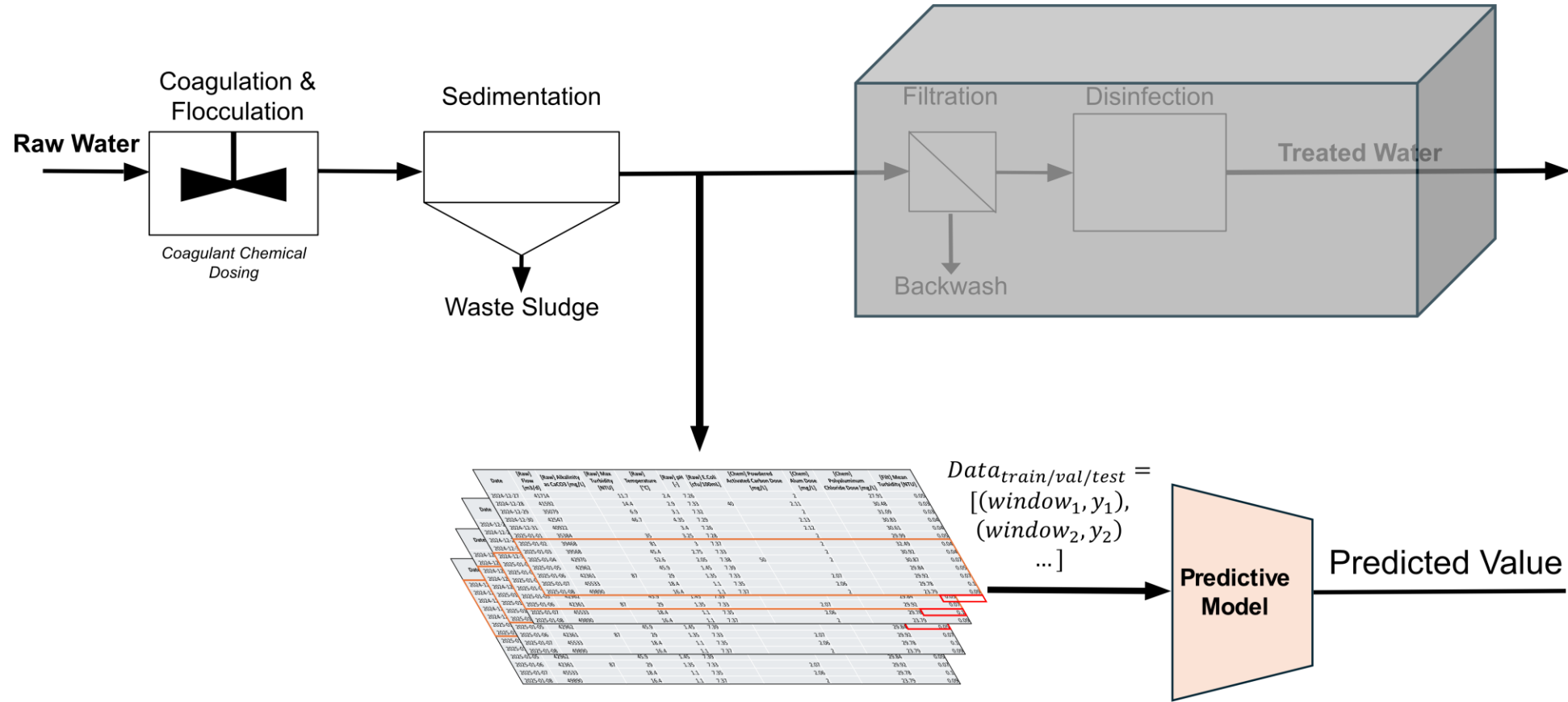
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2025-01-07	45533	87	0	1	18.4	1.1	7.35	50	0	3
2025-01-08	49890	87	0	2	16.4	1.1	7.37	50	0	4

Feature Engineering

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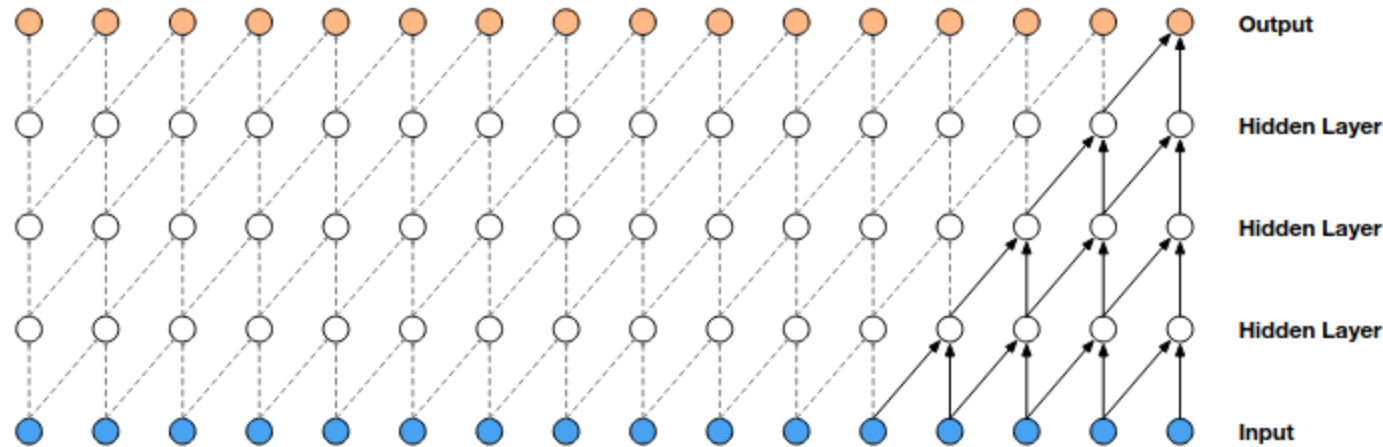




Models Explored

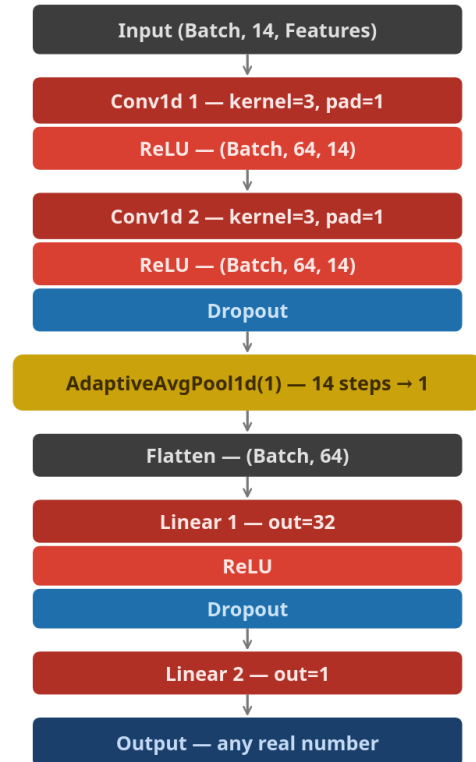
- **Dummy: Naïve (predict previous-day values)**
 - explore autoregressive nature of target variable
- **Baseline: XGBoost**
- **1D CNN**
 - Symmetric padding
 - Global average pooling
- **Temporal Convolutional Network**
 - *Pytorch-tcn*

Temporal-Convolutional Neural Network

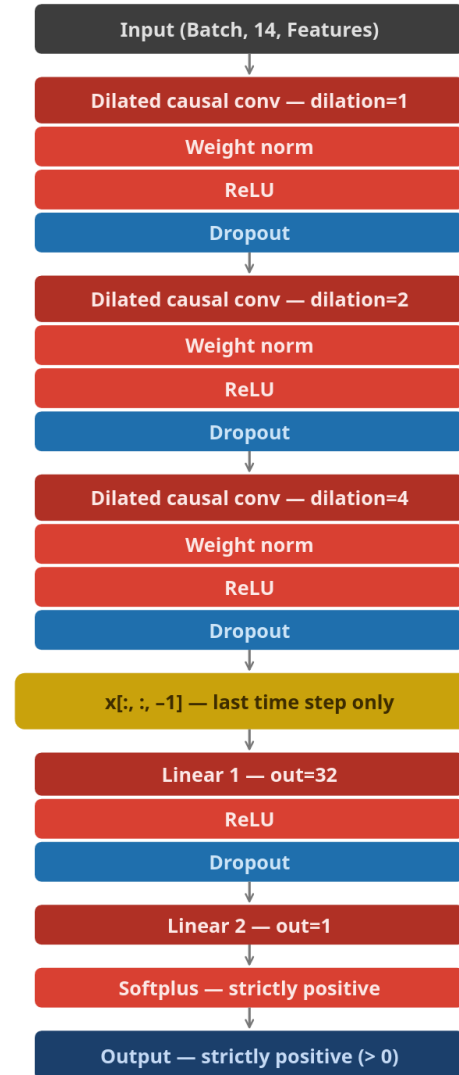


- **Causal Convolutions:** kernel only covers present & past data
- **Kernel:** input-weight dot product --> output feature map
 - Kernel size: 'window' size
- **Dilation:** increases receptive field, learn long-term dependencies
- **Stride:** dictates window overlap

Model 1 — CNN1DRegressor



Model 2 — TCNRegressor

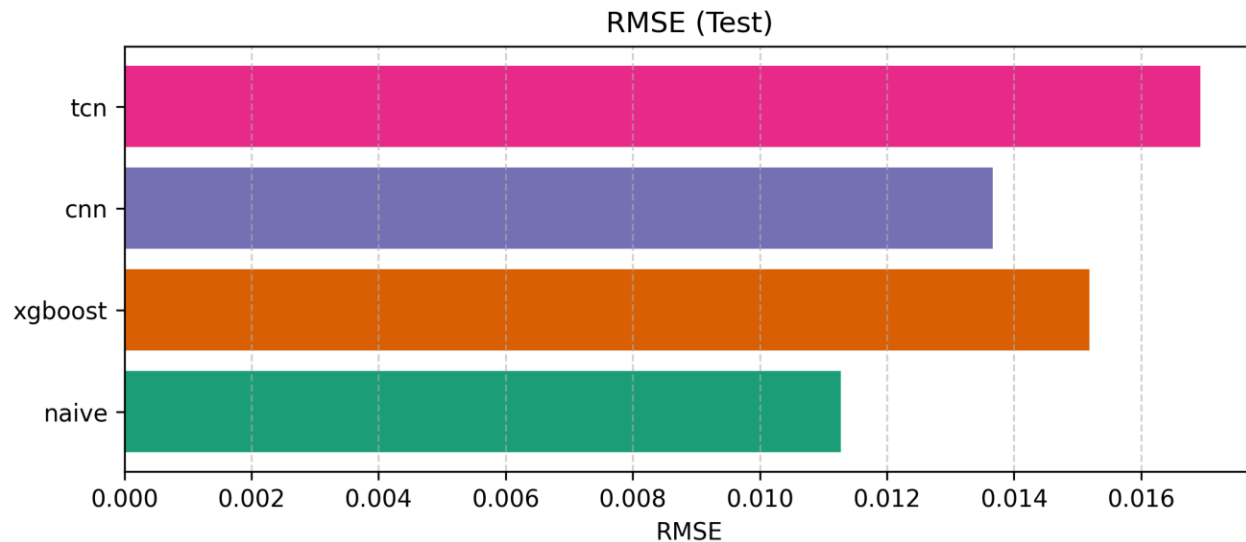
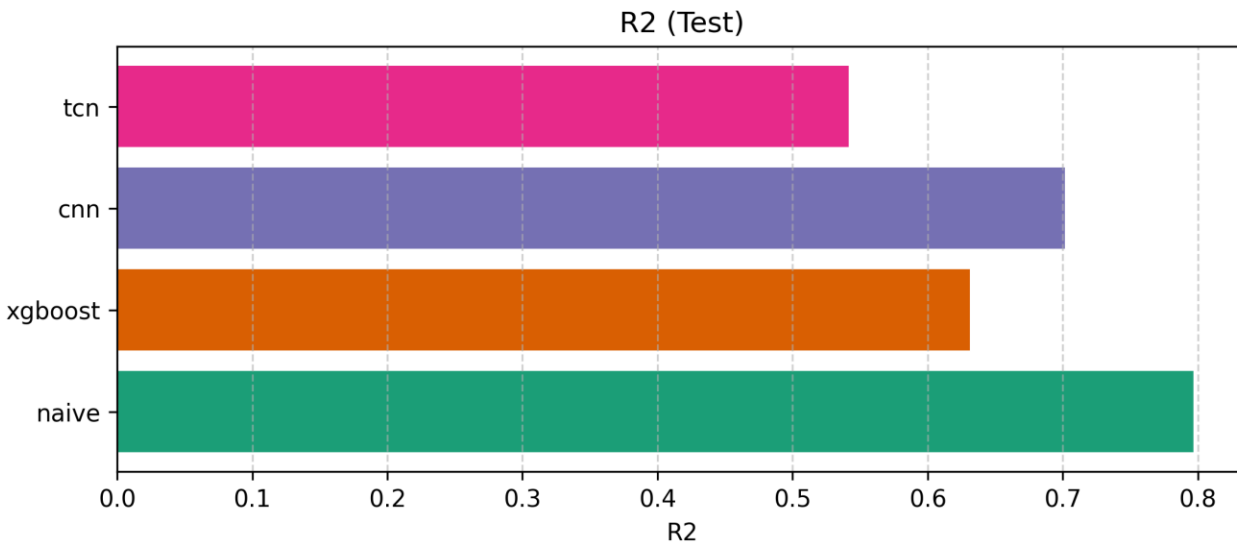


Raw data

- train/val/test split
- preprocessing
- sliding windows + targets
- numpy arrays + normalization
- build CNN
- train on X_{train} / y_{train}
- validate on X_{val} / y_{val}
- predict on X_{test}
- compare y_{pred} with y_{test}
- save metrics and predictions

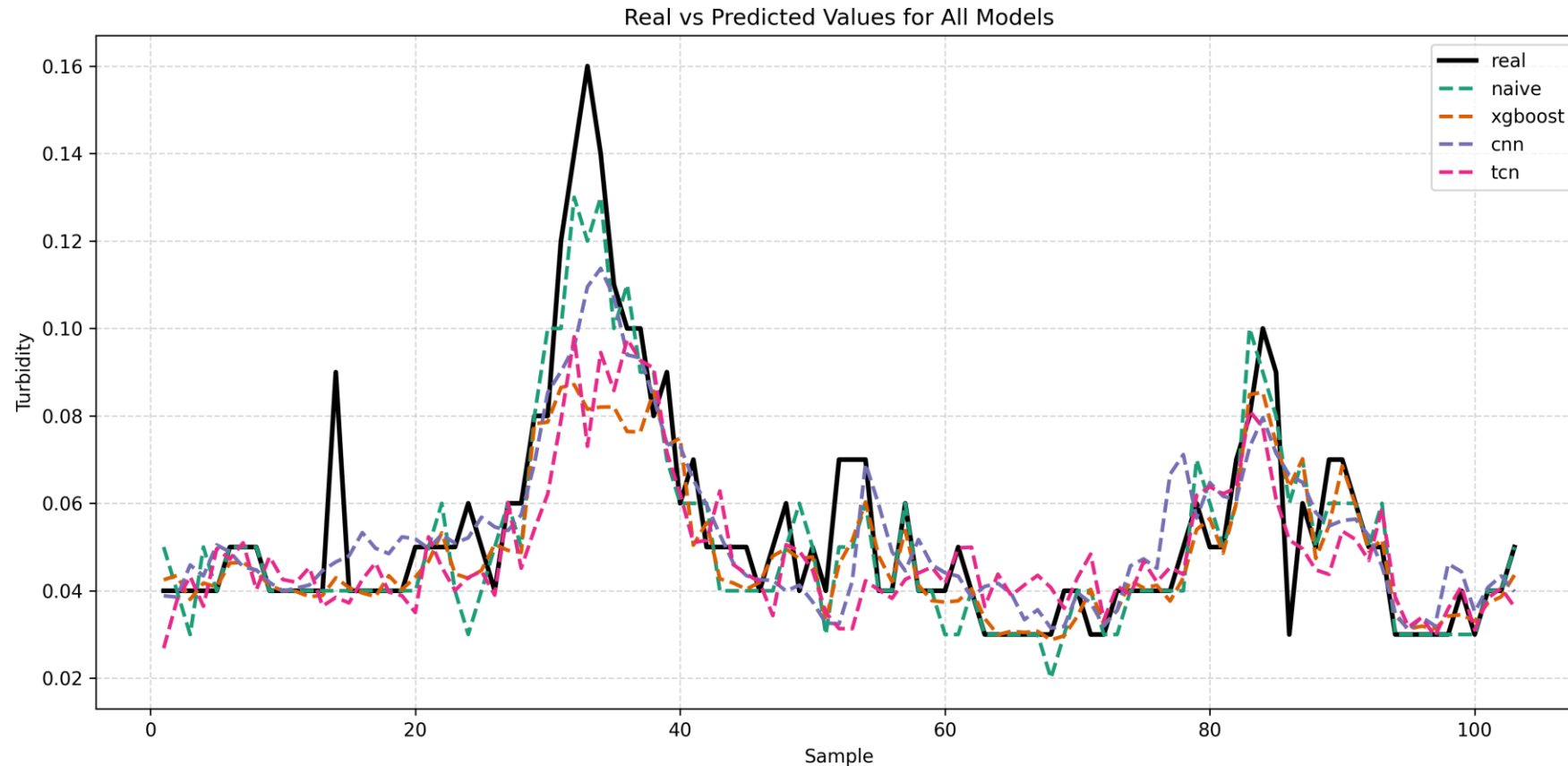
Model Results – Performance Metrics

Model Comparison on Test Set
Window = 14, Shift = 7



- Evaluated models on R^2 and RMSE to penalize missing high turbidity events.
- Models **underperformed** Naive baseline.
- Challenge in model generalization.

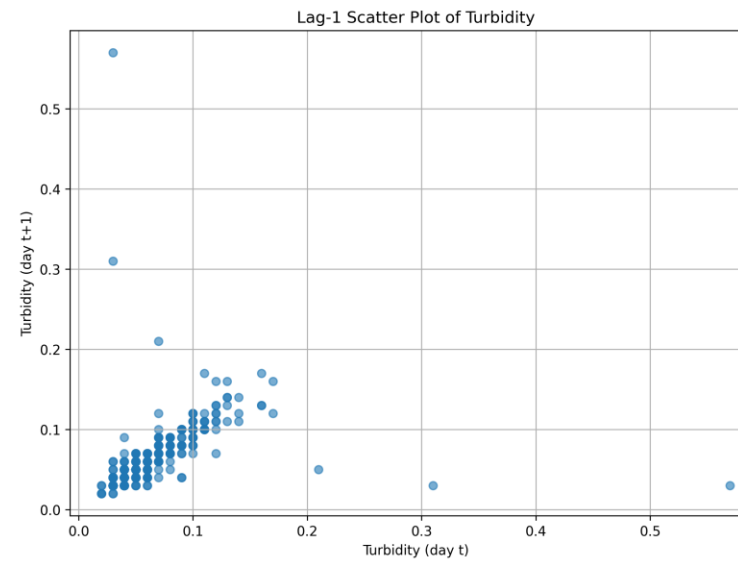
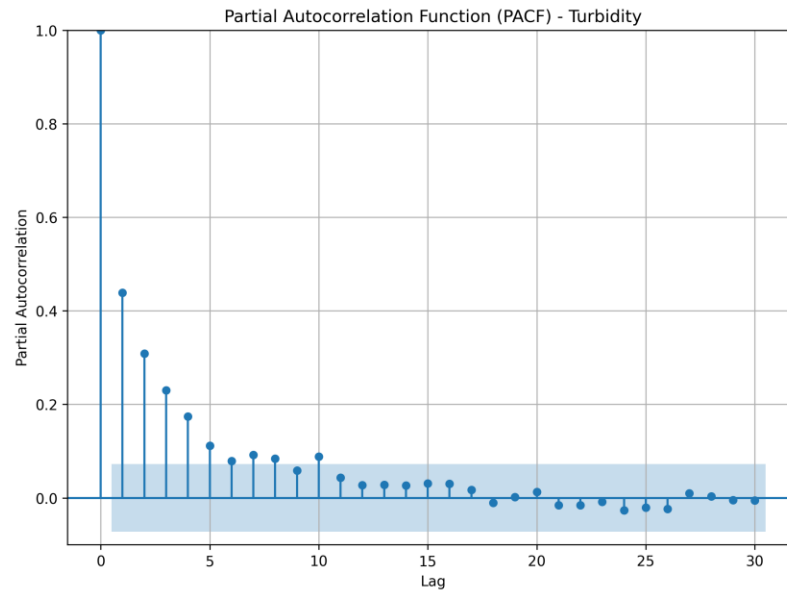
Model Test Results



- Models tend to **underestimate peak turbidity events**.
- *Most critical events to for safety and optimization!*

Understanding the Naïve Model

- Why naïve has the best performance
 - Turbidity exhibits **low variance** and **strong day-to-day persistence**
 - Behaves like a **short-memory autoregressive process**



Model Performance Is Limited by the Data

- **Model-data relationship**

- CNN captures local, short-term patterns
- XGBoost is well-suited for modeling structured, tabulated feature relationships
- TCN is designed to capture long-term temporal dependencies
- Model performance is **constrained by data properties**

- **Data limitation**

- **Variety** (e.g., suspended particle measurements or environmental factors), and
- Low data sampling **frequency**
- Shift step affects the structure of the dataset: small shift, windows overlap heavily; larger shift, we get fewer samples
- **Trade-off** between data quantity and data quality

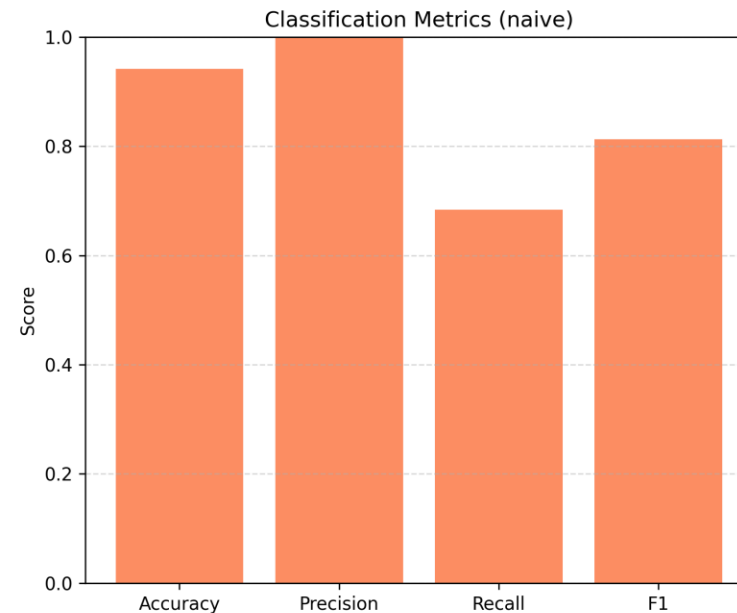
Attempts to Improve Models Performance

- **Model improvements**

- Additional feature engineering (e.g., dimensionality reduction)
- Testing different loss functions
- Hyperparameter tuning

- **Alternative Objective**

- Shifting from regression → **classification**



Models cannot compensate for limited data, performance is determined by the signal present in the data.

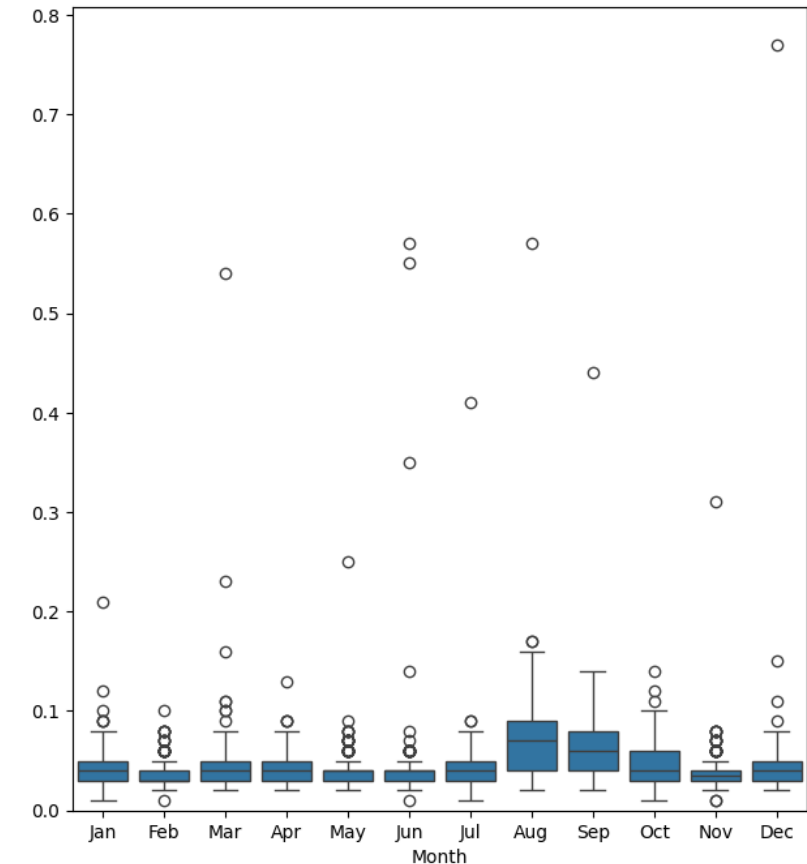
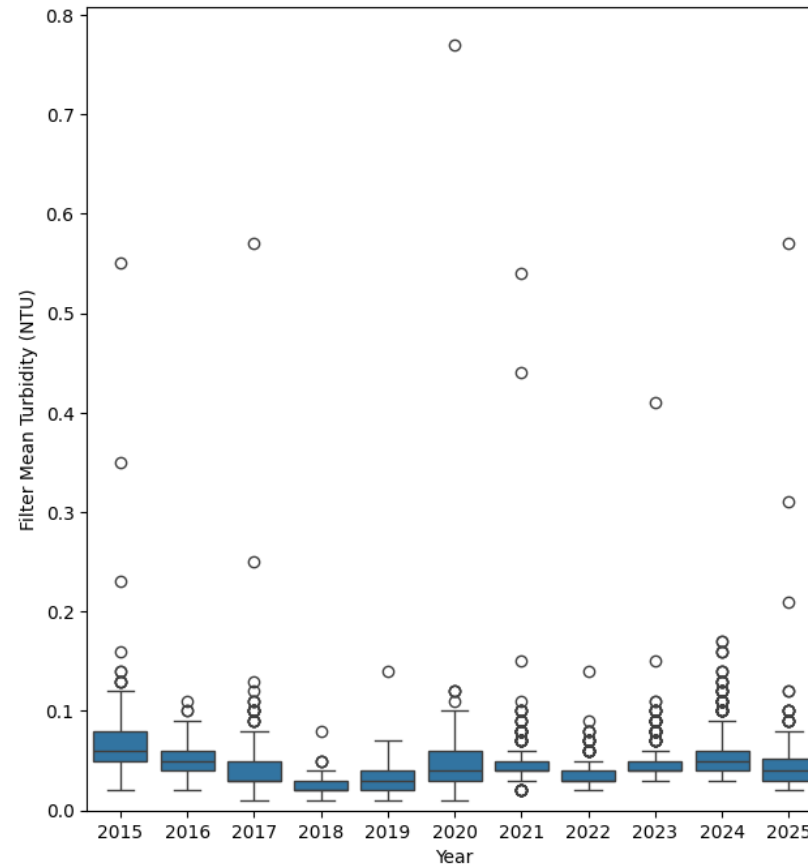
Thanks for your Attention!

Questions?

Appendix

Historical Turbidity Trend

- Mean and IQR varies from year to year: **data distribution changes.**
- **Seasonal trend** occurs: Higher turbidity observed in summer months.



Predicted vs Actual Comparison Plot

